1 Research Objectives and Physical Modeling

1.1 Project Context and Research Questions

This project investigates the vertical ascent of a variable-mass model rocket: how high it flies under quadratic drag, and whether fixed propellant should be burned rapidly or slowly. Analytical solutions, three explicit integrators and independent verification are used to answer these questions.

1.2 Parameters and Governing Equations

Physical quantity	**Symbol**	**Baseline**	**SI unit**
Initial / dry mass	m_0 / m_f	1.0 / 0.6	kg
Propellant / burn rate	Δm / q	0.4 / 0.2	kg / kg s^{-1}
Effective exhaust velocity	v_e	500	m s^{-1}
Drag coefficient	c	0 or 0.002	kg m^{-1}
Gravity	g	9.81	m s^{-2}
Thrust / burn duration	T / t_b	100 / 2	N / s

With upward positive, y=(h,v,m), h(0)=v(0)=0 and m(0)=m_0. The burn ends at t_b=(m_0−m_f)/q. The phase equations are:

$$h' = v, \qquad v' = \frac{T - cv|v|}{m} - g$$

$$0 \le t < t_b\colon\ T = qv_e, \qquad m' = -q; \qquad t \ge t_b\colon\ T = 0, \qquad m' = 0, \qquad m = m_f$$

Integration stops at the first post-burnout v=0 crossing (apogee). Initial lift-off requires $qv_e > m_0 g$; the baseline thrust-to-weight ratio is 10.194.

1.3 Assumptions and Dimensional Check

The model assumes vertical point-mass motion, constant g, v_e and c, a constant powered burn rate and instantaneous shutdown. It neglects changing atmospheric density, wind, rotation, motor transients and recovery. The mass is continuous at burnout, but acceleration is discontinuous.

Both qv_e and $cv|v|$ have units of force (N); division by m gives m/s^2. The signed drag term always opposes velocity. Numerical verification checks this mathematical model; it is not experimental validation of a physical rocket.

2 Analytical Solutions for the Frictionless Case

2.1 Powered Flight

For c=0, conservation of momentum gives m dv/dt=qv$_e$−mg and m=m$_0$−qt. Dividing by m and integrating from launch yields:

$$v(t) = v_e ln[\frac{m_0}{m_0 - qt}] - gt$$

A second integration, with the substitution m=m$_0$−qt, gives:

$$h(t) = \frac{v_e}{q}\{qt - (m_0 - qt)ln[\frac{m_0}{m_0 - qt}]\} - \frac{gt^2}{2}$$

At burnout, let L=ln(m$_0$/m_f). The exact event values are:

$$v_b = v_e L - g t_b$$

$$h_b = \frac{v_e}{q}[(m_0 - m_f) - m_f L] - \frac{g{t_b}^2}{2}$$

2.2 Coasting to Apogee

With τ=t−t_b, the mass is constant and acceleration is −g:

$$v = v_b - g\tau, \quad h = h_b + v_b\tau - \frac{g\tau^2}{2}, \quad m = m_f$$

$$t_{apogee} = t_b + \frac{v_b}{g}, \qquad H = h_b + \frac{{v_b}^2}{2g}$$

For the baseline, v_b=235.792812 m/s, h_b=214.141564 m, H=3047.895394 m and t_apogee=26.035965 s. These closed-form histories provide reference values at every exported time, not only at the two events.

2.3 Exact Relations for Coasting with Drag

For c>0 and upward coasting, v′=−g−cv²/m_f. Separating variables gives:

$$H - h_b = \frac{m_f}{2c} ln[1 + \frac{c{v_b}^2}{m_f g}]$$

$$t_{coast} = \sqrt{\frac{m_f}{cg}}\ arctan[v_b \sqrt{\frac{c}{m_f g}}]$$

Both expressions approach the drag-free limits as c→0. They provide additional checks; the independent DOP853 reference still integrates both flight phases. Detailed integration steps and the no-drag burn-rate argument are retained in Appendix A.

3 Numerical Methods and Order of Convergence

3.1 Time Integration and Event Handling

The project implements Forward Euler, Heun and classical RK4 directly. Euler uses one slope, Heun averages initial and predicted slopes, and RK4 combines four slopes. Their global orders on smooth intervals are 1, 2 and 4; formulas are given in Appendix B.

The final powered step is shortened to land exactly at burnout, and every stage uses the powered RHS. The next step uses the coasting RHS. After velocity changes sign, bisection evaluates fractional steps from the same left state until the bracket is $\leq 10^{-12}$ s, with a 100-iteration limit. This refines the numerical event but does not remove accumulated integration error.

3.2 Measured Convergence and Step Selection

$$E(\Delta t) = |Q_{num} - Q_{ref}|, \qquad p = log_2[\frac{E(\Delta t)}{E(\Delta t/2)}]$$

The local study uses $\Delta t = \{0.1, 0.05, 0.025, 0.0125\}$ s against the no-drag solution. A log–log fit gives the burnout-velocity orders below. wmlab measures the full state at the smooth fixed horizon t=1.5 s with $\Delta t = \{0.1, 0.05, 0.025\}$ s; the two studies therefore test different quantities.

Method	Theoretical order	Local fitted order	wmlab order
Euler	1	0.996296	0.988516
Heun	2	1.999932	1.999868
RK4	4	4.001140	3.999388

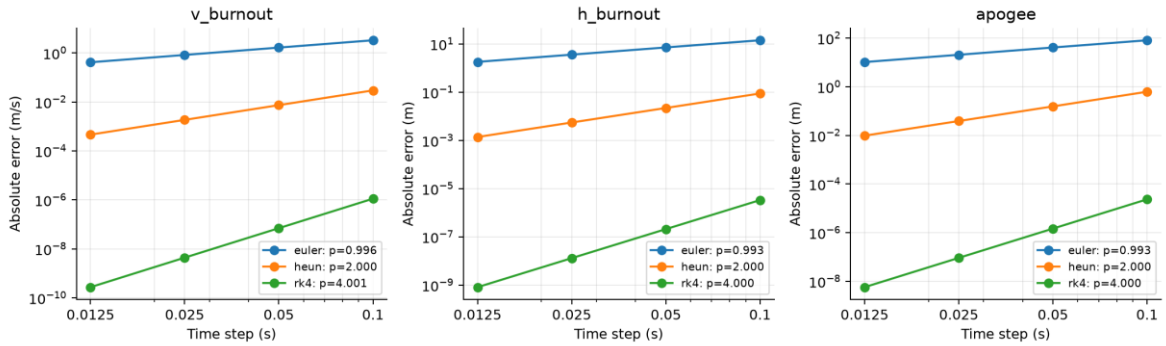


Figure 1. Local no-drag errors for burnout velocity, burnout altitude and apogee over four step sizes.

RK4 with maximum $\Delta t = 0.02$ s is used for baseline results: the independent trajectory errors are below 1.6×10^{-7} m and 6.7×10^{-8} m/s. Step selection is driven by accuracy; a failed coarse-step check alone is not proof of instability. Appendix B gives a local stability diagnostic.

Source: outputs/convergence.csv; convergence/wmlab/*_report.json.

4 Baseline Simulation Results and Peak Acceleration Analysis

4.1 Drag versus Drag-Free Ascent

The cases use identical baseline parameters and RK4 with maximum $\Delta t=0.02$ s; only c changes. Burnout occurs at 2 s in both cases.

Metric	No drag	With drag	Unit
Burnout velocity	235.793	172.272	m/s
Burnout altitude	214.142	183.593	m
Apogee altitude	3047.895	544.420	m
Time to apogee	26.036	8.999	s
Maximum upward a/g	15.989	9.841	1
Maximum |a|/g	15.989	11.084	1

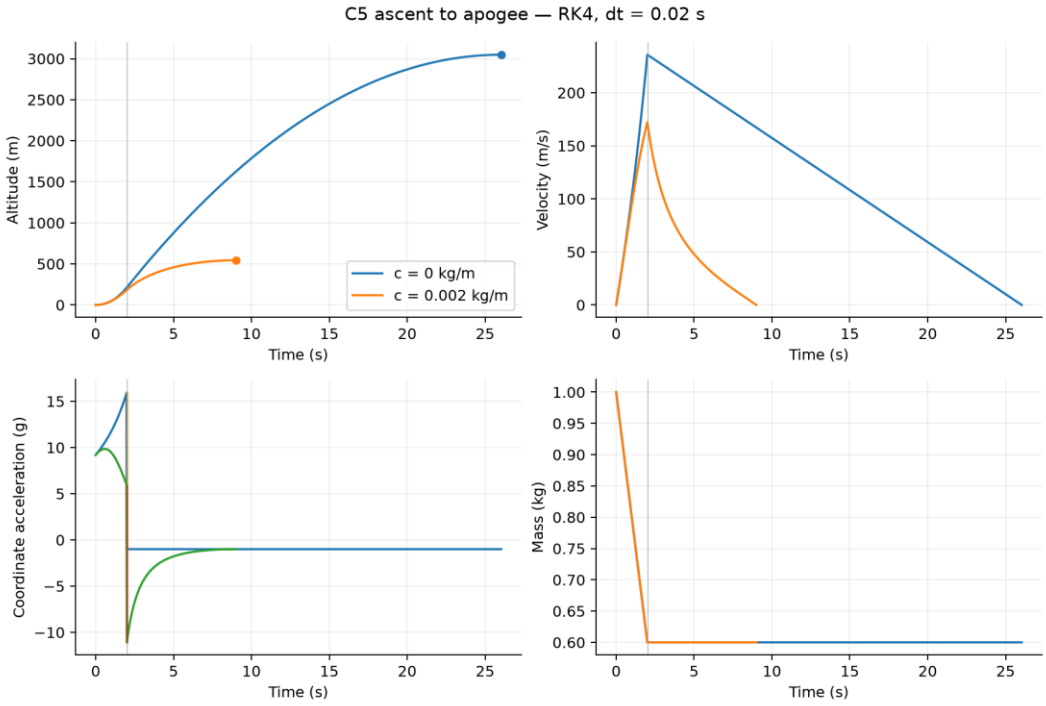


Figure 2. Baseline altitude, velocity, coordinate acceleration and mass. Burnout is at t=2 s.

4.2 Acceleration Interpretation

Drag lowers apogee by 82.14%. Its upward acceleration peaks near 0.58 s (9.841g), falling to 5.905g at t_b^- and jumping to −11.084g at t_b^+. Thus the largest |a| occurs after shutdown. Proper acceleration is a+g; peak |a+g|/g is 16.989 without drag and 10.841 with drag. These point-mass diagnostics do not establish structural stress.

Source: outputs/baseline.csv; outputs/figures/trajectories.png.

5 Burn Rate Sensitivity and Flight Optimization

5.1 Parameter Sweep and Bounded Refinement

For fixed propellant mass 0.4 kg, the study starts at $q_{min}=1.05m_0g/v_e=0.020601$ kg/s, a 5% lift-off thrust margin. Each sweep uses 65 logarithmic samples; upper bounds 0.5, 1 and 2 kg/s check domain sensitivity. Detected peaks are refined by bounded scalar minimization and compared with the endpoints.

$$c = 0.002\ kg/m\text{:}\ q\ast \approx 0.04509253\ kg/s, \qquad t_\beta \approx 8.871\ s, \qquad H \approx 642.411947\ m$$

This candidate gains 97.99 m (18.00%) over q=0.2 kg/s. Halving the integration step changes H by about 1.85×10^{-8} m; its DOP853 discrepancy is about 1.97×10^{-8} m. It is a numerical optimum within the investigated domain, not a universal global-optimality proof.

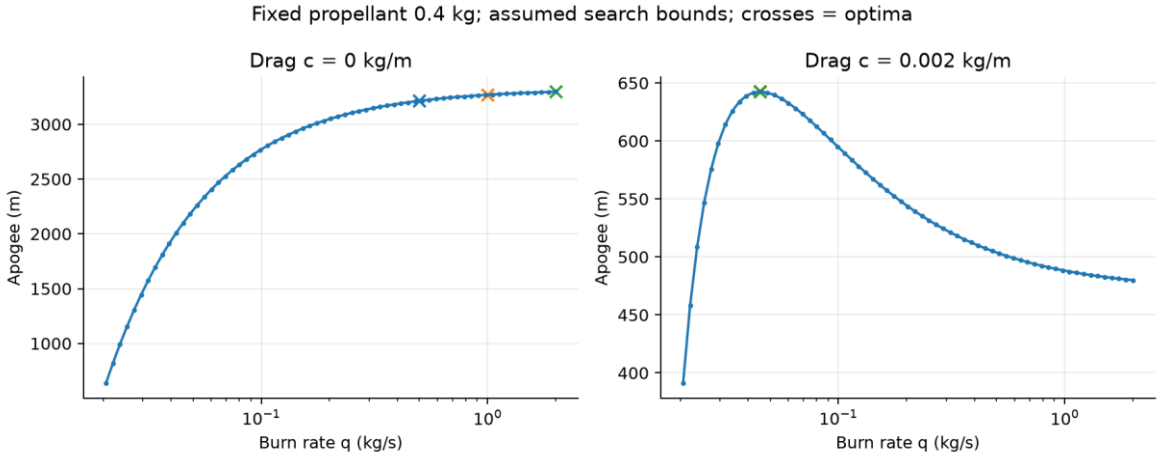


Figure 3. Apogee versus burn rate for fixed propellant mass, with and without quadratic drag.

5.2 Gravity–Drag Trade-Off

Low burn rates prolong powered flight and increase gravity losses. Higher rates shorten the burn but raise velocity, increasing drag force cv^2 and dissipation cv^3. The drag-inclusive maximum balances these effects. The constant c model does not represent atmospheric-density variation with altitude.

5.3 No-Drag Limit

Because $m_0L>m_0-m_f$, H increases monotonically with q toward $v_e^2L^2/(2g)$. The unconstrained no-drag model therefore has no finite maximizing burn rate. Motor or structural constraints would be needed for a realizable optimum.

Source: outputs/burn_rate_sweep.csv; outputs/burn_rate_optima.csv.

6 Verification Framework, Dataset Synthesis, and Limitations

6.1 DE Kit, wmlab and Independent Checks

The DE Starter Kit supplies the module, scenario and system-card templates. wmlab 0.1.1 loads scenario YAML and calls c5_rocket.py, which adapts the project simulator to the toolkit's t,Y interface. Euler, Heun and RK4 remain the project's own implementations. wmlab writes trajectory files and JSON check reports; its dataset command writes c5_rocket.npz.

Toolkit scenario	Method / Δt	Actual	Expected
c5_baseline_drag	RK4 / 0.02 s	PASS	PASS
c5_analytic_no_drag	RK4 / 0.02 s	PASS	PASS
c5_euler_coarse_fail	Euler / 0.5 s	FAIL	FAIL

The coarse Euler reference error is 0.313115, above 10^{-6} despite acceptable physical bounds: an accuracy failure, not proof of instability. The runner reads JSON verdicts because wmlab can exit with code 0 even when checks fail. Archived evidence records 172 passing pytest tests and three passing convergence reports.

Every exported h,v,m sample is compared at matching times with the analytical solution (c=0) or an independent, burnout-split DOP853 integration (c>0; rtol=atol=2×10^{-12}). Dynamics checks test $h'=v$, acceleration and phase-specific mass slopes with one-sided burnout treatment. The clean trajectories pass; copied-trajectory mutations are detected. Detailed tolerances, errors and mutation results are in Appendix C.

6.2 Dataset and System Card

wmlab generates 200 common-time ascent segments with seed 20261027 and stride=1. c5_rocket.npz stores Y with shape (200,201,3), ordered h,v,m, on 0–4 s at Δt=0.02 s. Phase-correct fractional steps are used instead of PCHIP. All 200 segments pass independent validation; maximum h/v/m errors are 2.69×10^{-7} m, 1.76×10^{-7} m/s and 1.40×10^{-14} kg. system_card.yaml records the authors, model, sampling assumptions and validity limits.

6.3 Scope and Conclusion

The results answer both project questions: baseline drag reduces apogee from 3047.90 to 544.42 m, while a slower burn near 0.04509 kg/s raises the drag-case apogee to 642.41 m. Verification supports the tested numerical model and exported data, not real-flight accuracy or absence of all possible faults. The model omits wind, rotation, variable density, transonic effects and motor transients. Reproduction details are in Appendix C and README.md.